a unified being because we reflect the world, because we have formed an image of a unified entity by interacting with our kind, and because it is a perspective on the world unified by memory (chapter 12). From this comes what we call the “flowing” of time. This is what we are listening to when we listen to the passing of time.

The variable “time” is one of many variables that describe the world. It is one of the variables of the gravitational field (chapter 4): at our scale, we do not register quantum fluctuations (chapter 5), hence it is possible to think of spacetime as determined, as Einstein’s great mollusk; at our scale, the movements of the mollusk are small and can be overlooked. Hence we can think of spacetime as being as rigid as a table. This table has dimensions: the one that we call space, and the one along which entropy grows, called time. In our everyday life we move at low speeds in relation to the speed of light and so we do not perceive the discrepancies between the different proper times of different clocks, and the differences in speed at which time passes at different distances from a mass are too small for us to distinguish.

In the end, therefore, instead of many possible times, we can speak only of a single time: the time of our experience—uniform, universal, and ordered. This is the approximation of an approximation of an approximation of a description of the world made from our particular perspective as human beings who are dependent on the growth of entropy, anchored to the flowing of time. We for whom, as Ecclesiastes[126] has it, there is a time to be born and a time to die.

This is time for us: a multilayered, complex concept with multiple, distinct properties deriving from various different approximations.

Many discussions of the concept of time are confused because they simply do not recognize its complex and multilayered aspect. They make the mistake of not seeing that the different layers are independent.

This is the physical structure of time as I understand it, after a lifetime of revolving around it.

Many parts of this story are solid, others plausible, others still are guesses hazarded in an attempt at understanding the whole.

Practically all the things recounted in the first part of the book have been ascertained from innumerable experiments: the slowing down of time according to altitude and speed; the nonexistence of the present; the relation between time and the gravitational field; the fact that the relations between different times are dynamic, that elementary equations do not recognize the direction of time; the relation between entropy and blurring. All this has been well ascertained.[127]